

DAY 1 : HDOC

Topic: Basic User Inputs, Operations & Output

1. Problem Statement

Q1 (User Inputs, Operations & Output)

Write a program to input two numbers and display their sum.

2. Algorithm

: Begin the execution of the program

: Declare three integer variables , number 1, number 2, and sum

: Read two integer inputs from the user using scanf.

: Calculate the addition using

Sum= num1+ num2 .

: Output the result on the console in the required format sum< value > .

: Terminate the program execution (return 0)

3. Test cases table

Test Case	1 st Input	2 nd Input	Expected Output
Test case 1	2	3	5
Test case 2	-1	20	19

4. C Program Source Code

```
#include <stdio.h>
```

```
int main() {
```

```
    int num1, num2, sum;
```

```
printf("Enter two numbers: ");  
  
scanf("%d %d", &num1, &num2);  
  
sum = num1 + num2;  
  
// Display result  
printf("Sum = %d\n", sum);  
  
return 0;  
}
```

```
1 #include <stdio.h>  
2  
3 int main() {  
4     int num1, num2, sum;  
5  
6     printf("Enter two numbers: ");  
7     scanf("%d %d", &num1, &num2);  
8  
9     // Calculate sum  
0     sum = num1 + num2;  
1  
2     // Display result  
3     printf("Sum = %d\n", sum);  
4  
5     return 0;  
6 }
```

- **EXECUTION OUTPUT :**

Output

Enter two numbers: 2 3

Sum = 5

=== Code Execution Successful ===

Output

Enter two numbers: -1 20

Sum = 19

=== Code Execution Successful ===

Topic: Arithmetic Operations & User Inputs in C

1. Problem Statement

Q2 (User Inputs, Operations & Output)

Write a program to input two numbers and display their sum , difference , product , and quotient.

Note: Division by zero should be handled by carefully.

2. Algorithm

Step 1: Start: Begin the program execution.

Step 2: Declare Variables: Declare integer or floating-point variables for two numbers (a, b) and

their results (sum, diff, product, quotient).

Step 3: Input Values: Read two numbers from the user using scanf().

Step 4: Compute Sum: Calculate $\text{sum} = a + b$.

Step 5: Compute Difference: Calculate $\text{diff} = a - b$.

Step 6: Compute Product: Calculate $\text{product} = a * b$.

Step 7: Compute Quotient: Check if $b \neq 0$. If true, calculate $\text{quotient} = a / b$. Otherwise, handle

division by zero.

Step 8: Display output

Step 9 : Stop

3 Test cases table

Test case	Input 1 (a)	Input 2 (b)	Expected Output
Test case 1	10	2	Sum=12 , diff =9 , Product = 20, Quotient = 5
Test case 2	7	3	Sum=10 , Diff = 4 , Product =21

4.C program source code

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b;
```

```
    int sum, diff, product, quotient;
```

```
    // Read two numbers from user
```

```
    scanf("%d %d", &a, &b);
```

```
    // Perform arithmetic operations
```

```
    sum = a + b;
```

```
    diff = a - b;
```

```
    product = a * b;
```

```
    // Check division by zero and display output
```

```
    if (b != 0) {
```

```
        quotient = a / b;
```

```
    printf("Sum=%d, Diff=%d, Product=%d,  
Quotient=%d\n", sum, diff, product, quotient);  
  
    } else {  
  
        printf("Sum=%d, Diff=%d, Product=%d,  
Division by zero error!\n", sum, diff, product);  
  
    }  
  
    return 0;  
  
}
```

```
1 #include <stdio.h>  
2  
3 int main() {  
4     int a, b;  
5     int sum, diff, product, quotient;  
6  
7     // Read two numbers from user  
8     scanf("%d %d", &a, &b);  
9  
10    // Perform arithmetic operations  
11    sum = a + b;  
12    diff = a - b;  
13    product = a * b;  
14  
15    // Check division by zero and display output  
16    if (b != 0) {  
17        quotient = a / b;  
18        printf("Sum=%d, Diff=%d, Product=%d, Quotient=%d\n", sum, diff, product, quotient);
```

```
19     } else {  
20         printf("Sum=%d, Diff=%d, Product=%d, Division by zero error!\n", sum, diff, product);  
21     }  
22  
23     return 0;  
24 }
```

5. Program Execution Output

Execution output 1:

```
10 2  
Sum=12, Diff=8, Product=20, Quotient=5  
  
=== Code Execution Successful ===
```

Execution output 2:

7 3

Sum=10, Diff=4, Product=21, Quotient=2

=== Code Execution Successful ===